Type MS · Name the food-test reagent or read food-test results

Answer 1 of 12 · Question 3 part b.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
3	(a)	(i)	B: A, C and D are not correct as bile is not produced in the gall bladder, small intestine or stomach.		1
		(ii)	Three from: • emulsifies/emulsification fat/lipid; • increases surface area (of fat/lipid); • for lipases/enzymes to digest/work; • neutralises the contents from the stomach;		3
	(b)	(i)	• it contains starch; • it contains protein; • it does not contain reducing sugars/glucose;		3
		(ii)	• contains amylase/carbohydrase; • digested/breaks down the starch; • into glucose/maltose;		3
	(c)		Five from: • peristalsis; • of the food bolus; • (circular) muscles (behind the bolus) contract to push it along; • fibre is indigestible/not broken down; • adds bulk; • stimulates muscles to contract;		5

Total 15 marks

Type MS · Name the food-test reagent or read food-test results

Answer 1 of 12 · Question 3 part b.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 2 of 12 · Question 2 part b.iv · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

Question number				
2	(a)	(i)	W large intestine; X oesophagus; Y stomach; Z small intestine;	Allow colon Allow ileum	4
		(ii)	In the following order only: • bolus; • muscles; • contract; • peristalsis;		4
	(b)	(i)	salivary gland/ pancreas;		1
		(ii)	(lining of) **small** intestine/ ileum/ duodenum;		1
		(iii)	• starch digested/broken down to sugar/glucose; • (sugar) diffuses/moves out through (Visking tubing);	Allow maltose	2
		(iv)	Benedict's solution/reagent;		1
		(v)	Any two from the following: • blue/black colour; • (undigested) starch is present; • too large to pass through Visking tubing;		2

Total marks 15

Type MS · Name the food-test reagent or read food-test results

Answer 2 of 12 · Question 2 part b.iv · 1 mark

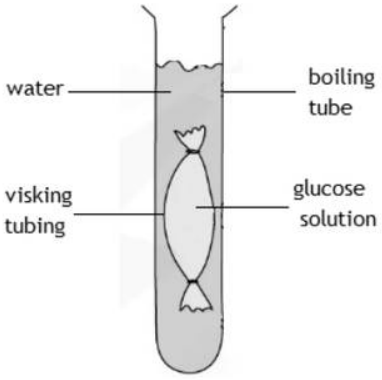
2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 3 of 12 · Question 5 part b.i · 1 mark

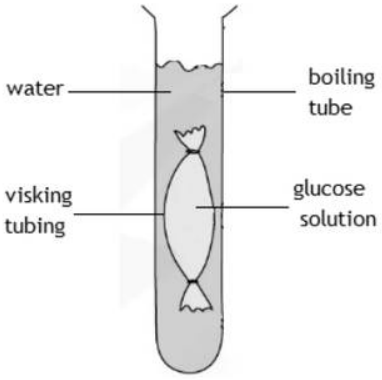
2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

5 (a)		• greater concentration gradient/high concentration in small intestine; • (more) glucose collides/in contact with, wall/epithelium, of small intestine; • faster rate of diffusion;	allow reverse argument	3
(b)	(i)	Benedict's (solution);		1
	(ii)	(heated) water bath/Bunsen burner/measuring cylinder;		1
	(iii)	• diagram; • labels;;  	1 mark for each set of two correct labels. ignore any inclusion of a water bath in diagram accept sugar solution and glucose/sugar solution	3
(c)`	(i)	villus/villi		1
	(ii)	any three from • many/large number of villi/have microvilli; • increase surface area; • thin walls; • for shorter diffusion distance; • capillaries; • to maintain diffusion/concentration gradient;	explanation must be linked to feature	3
				Total for Question 5 = 12 marks

Type MS · Name the food-test reagent or read food-test results

Answer 4 of 12 · Question 5 part b.ii · 1 mark

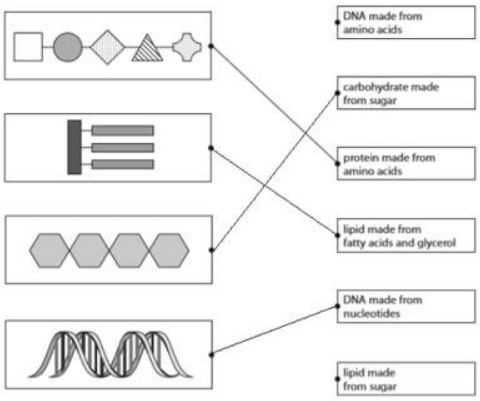
2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

5 (a)		• greater concentration gradient/high concentration in small intestine; • (more) glucose collides/in contact with, wall/epithelium, of small intestine; • faster rate of diffusion;	allow reverse argument	3
(b)	(i)	Benedict's (solution);		1
	(ii)	(heated) water bath/Bunsen burner/measuring cylinder;		1
	(iii)	• diagram; • labels;;  	1 mark for each set of two correct labels. ignore any inclusion of a water bath in diagram accept sugar solution and glucose/sugar solution	3
(c)	(i)	villus/villi		1
	(ii)	any three from • many/large number of villi/have microvilli; • increase surface area; • thin walls; • for shorter diffusion distance; • capillaries; • to maintain diffusion/concentration gradient;	explanation must be linked to feature	3
			Total for Question 5 = 12 marks	

Type MS · Name the food-test reagent or read food-test results

Answer 5 of 12 · Question 3 part c.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

3	(a)		Do not allow more than one line from each structure	3
	(b)	**D** stomach;		1
	(c) (i)	3/three		1
	(ii)	not washing equipment/transfer of food containing protein into distilled water;	Allow any valid alternative	1
	(d)	Hazard: Broken glass - cuts Reducing risk: Clear up breakages using a dustpan and brush/keep equipment away from edge of tables;	Ignore wear gloves	1
		Hazard: Biuret reagent - irritant Reducing risk: Wear goggles/wash hands after use/don't use if skin is sensitive;	Ignore wear gloves	1

Total question 3 = 8 marks

Type MS · Name the food-test reagent or read food-test results

Answer 5 of 12 · Question 3 part c.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 6 of 12 · Question 2 part b.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a)	4 of • add powder to water (in a test tube); • add Benedict's reagent/solution; • heat tube in a water bath; • observe any colour change/description of colour change; • wear goggles;		4 max
(b) (i)	powder Y;		1
(ii)	• powder W; • powder Z;		2
(iii)	powder X;		1
(iv)	protein;		1
			Total 9

ALWAYS LEARNING PEARSON

Type MS · Name the food-test reagent or read food-test results

Answer 6 of 12 · Question 2 part b.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 7 of 12 · Question 2 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a)	4 of • add powder to water (in a test tube); • add Benedict's reagent/solution; • heat tube in a water bath; • observe any colour change/description of colour change; • wear goggles;		4 max
(b) (i)	powder Y;		1
(ii)	• powder W; • powder Z;		2
(iii)	powder X;		1
(iv)	protein;		1
			Total 9

ALWAYS LEARNING PEARSON

Type MS · Name the food-test reagent or read food-test results

Answer 7 of 12 · Question 2 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 8 of 12 · Question 2 part b.iii · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

2	(a)	4 of • add powder to water (in a test tube); • add Benedict's reagent/solution; • heat tube in a water bath; • observe any colour change/description of colour change; • wear goggles;		4 max
	(b) (i)	powder Y;		1
	(ii)	• powder W; • powder Z;		2
	(iii)	powder X;		1
	(iv)	protein;		1
				Total 9

ALWAYS LEARNING PEARSON

Type MS · Name the food-test reagent or read food-test results

Answer 8 of 12 · Question 2 part b.iii · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 9 of 12 · Question 2 part b.iv · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b.iv in this mark scheme extract

2	(a)	4 of • add powder to water (in a test tube); • add Benedict's reagent/solution; • heat tube in a water bath; • observe any colour change/description of colour change; • wear goggles;		4 max
	(b) (i)	powder Y;		1
	(ii)	• powder W; • powder Z;		2
	(iii)	powder X;		1
	(iv)	protein;		1
				Total 9

ALWAYS LEARNING PEARSON

Type MS · Name the food-test reagent or read food-test results

Answer 9 of 12 · Question 2 part b.iv · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 10 of 12 · Question 2 part b.v · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.v in this mark scheme extract

Question number				
2	(a)	(i)	W large intestine; X oesophagus; Y stomach; Z small intestine;	Allow colon Allow ileum	4
		(ii)	In the following order only: • bolus; • muscles; • contract; • peristalsis;		4
	(b)	(i)	salivary gland/ pancreas;		1
		(ii)	(lining of) **small** intestine/ ileum/ duodenum;		1
		(iii)	• starch digested/broken down to sugar/glucose; • (sugar) diffuses/moves out through (Visking tubing);	Allow maltose	2
		(iv)	Benedict's solution/reagent;		1
		(v)	Any two from the following: • blue/black colour; • (undigested) starch is present; • too large to pass through Visking tubing;		2

Total marks 15

Type MS · Name the food-test reagent or read food-test results

Answer 10 of 12 · Question 2 part b.v · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.v in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 11 of 12 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	Iodine (solution);		1
(b)	protein;		1
(c)	• add Benedict's/solution B to food in a test/boiling tube; • heat; • using a water bath; • wear goggles;		4
			Total 6

Type MS · Name the food-test reagent or read food-test results

Answer 11 of 12 · Question 4 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the food-test reagent or read food-test results

Answer 12 of 12 · Question 4 part b · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	Iodine (solution);		1
(b)	protein;		1
(c)	• add Benedict's/solution B to food in a test/boiling tube; • heat; • using a water bath; • wear goggles;		4
			Total 6

Type MS · Name the food-test reagent or read food-test results

Answer 12 of 12 · Question 4 part b · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks